DISTRIBUTED EDGE FLISR

A student-friendly revision guide to the paper

A Distributed Edge FLISR Solution & Network Simulation Test Platform

---

Paper: arXiv:2407.17497v1
Link: https://arxiv.org/html/2407.17497v1
Purpose: Understand the core ideas, diagrams, results, and how to adapt them into a third-year EEE project.

THE BIG IDEA

A distribution grid should not only detect a fault and switch off power. It should also identify the affected section, isolate it, and restore electricity to healthy sections when a safe alternative supply path exists.

How to use these notes

- Read Sections 1–4 to understand the problem and the FLISR process.
- Use the diagrams in Sections 5–6 to revise the architecture and a possible student prototype.
- Use the results table and quick-revision sheet before a presentation or viva.

Important: These notes summarize the paper and add clearly labelled simplified examples for learning. The proposed ESP32 circuit and project architecture are adaptations, not circuit diagrams supplied by the paper.

1. What problem is the paper solving?

Electricity distribution networks contain feeders, branches, switches, protection devices, and customer loads. When a fault occurs, protection may disconnect a larger area than the faulted section itself. Healthy customers can therefore lose supply even though their section is not damaged.

Traditional response may involve locating the fault, sending a crew, operating switches, and restoring service manually. Automated FLISR aims to make this process faster and reduce the number of customers and the duration of interruptions.

A simple feeder example

```
Substation → S1 → Section A → S2 → Section B → S3 → Section C → Loads
```

If Section B develops a fault, the goal is to isolate B. If the network has a suitable alternate supply path, healthy Section C—or another healthy section—may be restored without energizing the fault.

Key terms

Term	Meaning	Why it matters
Feeder	A distribution circuit supplying downstream sections.	Defines the route along which power flows.
Fault	An abnormal electrical condition, such as a short circuit or earth fault.	May trigger protection and interrupt supply.
Switch / breaker	A device used to connect or disconnect part of a circuit.	Used to isolate the fault or reconfigure the network.
Topology	How sources, lines, switches, and loads are connected.	Determines which restoration routes are possible.
Self-healing grid	A grid that detects problems and automatically takes corrective actions.	Can reduce outage area and restoration time.

Important distinction

Fault detection is not the same as fault location. An overcurrent sensor may tell you that something abnormal happened, but additional measurements, indicator states, and knowledge of the network are needed to infer which section is faulty.

2. FLISR: the central concept

Step	Meaning	Typical action
F — Fault	Detect an abnormal condition.	Use protection status, current, voltage, or an event flag.
L — Location	Identify the likely faulted section.	Combine indicators with the known network topology.
I — Isolation	Disconnect the faulted section.	Open selected switches and verify their state.
SIR — Service Restoration	Restore healthy sections if safe and possible.	Close a suitable tie switch or reconfigure the network.

Memory aid: detect → locate → isolate → check the alternate path → restore healthy loads.

3. Centralized vs distributed FLISR

Centralized FLISR

```
Sensors / switch status → Communication network → Central controller
Central controller → Commands → Field switches
```

- Field devices send measurements and status to a central controller.
- The central controller evaluates the network and issues switching commands.
- Benefit: a central view can coordinate decisions across a wide network.
- Limitation: communication delay or loss can delay or prevent central decisions.

Distributed edge FLISR

```
Local measurements + switch status → Edge controller → Local FLISR decision
↕
Central system when available
```

- An edge controller runs decision logic near the equipment.
- Local decisions can reduce dependence on a round trip to the control center.
- The design still needs trustworthy local information, safe switching rules, and suitable protection.

Why edge computing matters

The paper investigates how local FLISR can operate with less dependence on communication availability and latency. Its industrial interfaces and software architecture are more advanced than a student prototype needs, but the principle can be tested on a small feeder model.

4. Signals and indicators

Signal	What it indicates	Caution
FPI Fault Passage Indicator	Fault current has passed a monitored point, according to the device's logic.	Does not automatically give the exact physical fault location.
LVI Loss of Voltage Indicator	Voltage has been lost at a monitored point.	Loss of voltage can result from upstream switching or supply loss, not only a local fault.
Switch status	Whether a switch is open or closed, as reported or sensed.	A command is not proof that a switch actually operated; feedback is valuable.
Topology data	Connectivity and relationships among network sections and switches.	Incorrect topology can produce unsafe restoration decisions.

The paper describes status messages involving FPI and LVI information, network/switch state, and an edge device that interacts with existing control equipment. Do not assume every implementation uses the same sensors or exact logic.

Communication terms mentioned in the paper

- **RTU:** Remote Terminal Unit, used to monitor and control remote equipment.
- **MQTT:** A lightweight publish/subscribe messaging protocol.
- **IEC 60870-5-104:** A communication protocol used in power-system telecontrol.

These are industrial concepts to understand, not mandatory components for a low-cost student build.

5. Diagrams and figures worth revisiting

The paper's figures are mainly network topologies, software architecture, simulation interfaces, and results—not a ready-made electronics circuit schematic.

Figure	What to look for	Study question
9–11	Example distribution-network topologies (Waterford, Kerry, Mullingar).	Where are the sources, branches, switches, and alternate paths?
12	Edge-device interfaces and interactions.	How does the edge FLISR connect to the field/control system?
13	Simulation platform interface.	How is the network represented to the operator?
14	Example fault test case.	How is a fault selected or represented in the simulator?
15	Simulation interfaces and interactions.	How do status and control messages move through the platform?
16	Example simulation results.	What changes after the algorithm operates?

Open the original paper to inspect these figures at full size: https://arxiv.org/html/2407.17497v1

6. A possible student prototype (adaptation, not the paper's circuit)

Suggested title: Low-Cost Distributed FLISR for a Smart Distribution Network with Hardware-in-the-Loop Fault Simulation.

Functional block diagram

```
Low-voltage isolated supply
↓
Feeder sections + loads + switches
↓
Current / voltage sensing
↓
Signal conditioning / ADC inputs
↓
ESP32: fault logic + network state
↓
Isolation + safe restoration logic
↓
Relay drivers → low-voltage switches
↓
Load indicators + LCD / dashboard
```

Suggested physical setup

- Use an isolated, current-limited low-voltage DC supply.
- Represent each feeder section with wiring, a controllable switch, and an LED/resistor or other low-voltage load.
- Measure current and voltage at selected points; use properly scaled and protected sensor outputs for the ESP32 ADC.
- Use relay-driver modules or suitable transistor drivers; never drive a relay coil directly from a microcontroller GPIO unless the board explicitly supports it.
- Add switch-state feedback where possible, so the controller can check whether a command succeeded.
- Use LEDs for customer/load status and a display or dashboard for fault location and switching state.

What is not specified here

This is a functional architecture, not a pin-by-pin schematic or validated bill of materials. Exact parts, ratings, wiring, ADC scaling, isolation, and protection depend on your chosen hardware. Do not connect an experimental circuit directly to mains or a real distribution feeder.

7. Fault demonstration: physical, virtual, or hybrid?

Method	How it works	Strength / limitation
Physical	Create a controlled abnormal condition on a low-voltage test feeder.	Shows real sensor values and switching, but has a limited set of safe scenarios.
Virtual	Inject a fault event into software and update the simulated network state.	Can test complex or rare cases without building dangerous fault conditions; does not prove real sensing.
Hybrid	Use real sensor/switch hardware plus virtual faults or communication events.	Good balance of physical demonstration and many repeatable test cases.

Recommended: hybrid testing. Label every result clearly as measured on hardware, simulated, or generated by software. The paper's simulator illustrates that a fault can be represented as a change in switch/network state without creating a real short circuit.

8. Results reported by the paper

The following are reported examples from the paper's specific test scenarios. They are not guaranteed results for every network or implementation.

Reported example	Before	After	Interpretation
Waterford scenario: affected customers	1,370	171	About 87.5% fewer customers affected in that case.
Customer-minutes-lost rate, as reported per hour	78,090	9,747	The interruption impact was substantially reduced in that scenario.

Reported operation-time comparison

Method / network	Average operation time reported
5G centralized	212.8 ms
4G LTE centralized	260.4 ms
3G centralized	267.1 ms
2G centralized	1,049.6 ms
Distributed FLISR	Approximately 2 ms

Interpret carefully: these are timings in the authors' test setup and measurement definitions. They should not be presented as universal FLISR response times. A local algorithm's timing may exclude sensor delay, breaker operating time, communication to other devices, and safety checks.

A limitation to remember

A restoration algorithm cannot restore every load after every fault. The result depends on the fault location, available alternate paths, switch placement, load limits, source capacity, protection coordination, and network topology. Multiple faults can leave more customers disconnected.

9. How to turn the paper into your own project

Project objective

Build a small distribution-feeder demonstrator that detects a fault, isolates the affected section, and restores healthy loads when an alternate path is available. Compare local/distributed logic with centralized logic under communication delay and failure.

Suggested experiments

Experiment	What to measure / compare
1. Baseline: no FLISR	How many loads remain disconnected and for how long?
2. Centralized FLISR	Restoration time, affected loads, and dependence on communication.
3. Distributed FLISR	Local decision time and restoration success.
4. Add communication delay	How much does centralized response time increase?
5. Lose communication	Can local logic still operate safely with the information it has?
6. Multiple faults / unavailable tie switch	Which loads cannot be restored, and why?
7. Switch fails to operate	Does feedback reveal the failure and prevent unsafe follow-on actions?

Useful metrics

- **Fault-detection time:** time from injected/measured abnormality to fault declaration.
- **Isolation time:** time until the faulted section is confirmed disconnected.
- **Restoration time:** time from fault event to restored healthy loads.
- **Loads interrupted:** number of loads that lose supply.
- **Loads restored:** number of healthy loads successfully re-energized.
- **Communication dependence:** performance with delay, packet loss, or outage.
- **Safety checks:** no restoration path may energize the faulted section or improperly connect sources.

Possible project extension

Add simple load-capacity or voltage-drop constraints, or couple the controller to a MATLAB/Simulink or Python network model. This makes the analysis stronger than a relay-only demonstration.

10. Quick revision sheet

Question	Short answer
What does FLISR do?	Fault detection, location, isolation, and service restoration.
Why isolate a fault?	Prevent the faulted section from being energized and limit outage impact.
Why restore service?	Healthy disconnected loads may be supplied through a safe alternate path.
Centralized vs distributed?	Centralized decisions rely on a central controller and communications; distributed edge logic can make local decisions.
What is an FPI?	A fault-passage indicator; useful evidence, not necessarily an exact fault locator.
What is an LVI?	A loss-of-voltage indicator; it signals supply loss at a monitored point.
Does FLISR always restore every load?	No. Network topology, fault locations, capacity, switch availability, and safety constraints limit restoration.
Does the paper give a student circuit?	Its focus is the edge FLISR solution and simulation platform, not a ready-to-build ESP32 circuit.

11. Suggested reading checklist

- Read the Introduction: identify the problem and claimed contribution.
- Study Figures 9–11: trace sources, switches, and alternate paths.
- Study Figure 12: identify edge device, field interface, and central-system links.
- Study Figures 13–16: understand fault injection, message flow, and results.
- Check the Results and Future Work sections: note what was tested and what remains limited.

Source paper

A Distributed Edge FLISR Solution & Network Simulation Test Platform
arXiv:2407.17497v1
https://arxiv.org/html/2407.17497v1

These notes are a learning aid, not a substitute for the paper. Verify exact figures, terminology, units, and experimental conditions against the original text before using numbers in a formal report.